Which Way Does the Condition Point?

AP Statistics · Unit 2 · Topic 2.6 · B: 2026-10-26 · E: 2026-11-13 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Enduring Understanding VAR-4** — The likelihood of a random event can be quantified.
- **LO VAR-4.D** — Calculate conditional probabilities. [Skill 3.A]
- **EK VAR-4.D.1** — The probability that event A will occur given that event B has occurred is called a conditional probability and denoted $P(A \mid B) = P(A \cap B) / P(B)$.
- **EK VAR-4.D.2** — The multiplication rule states that the probability that events A and B both will occur is equal to the probability that event A will occur multiplied by the probability that event B will occur, given that A has occurred. This is denoted $P(A \cap B) = P(A) \cdot P(B \mid A)$.

Essential question: Why can reversing the condition produce a very different probability, even when both probabilities use the same table?

DOK-3 skill: 4.B · **FRQ pattern:** `conditional-direction-confusion-p-a-given-b-vs-b-given-a`

DOK rationale: Part (c) coordinates two conditional probabilities with the base-rate counts, adjudicates a diagnostic claim, and diagnoses why reversing the conditioning event changes the relevant denominator.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to identify the group behind the 90 percent before confirming a claim.
- Begin the combined 4.3–4.5 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require each conditional probability to name its denominator group.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose a claim and cite one cell or margin from the screening table.
- Complete the follow-along about conditional probability and the general multiplication rule.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Who belongs in the denominator after we already know the result was positive?
- What group supplies the denominator for the 90 percent sensitivity?
- How can a 10 percent false-positive rate create 95 false positives here?
- Which two conditional statements use the same intersection but different margins?

Adult role

Solo teacher. Circulate 5 pairs. Ask students to say each conditional probability in words and point to its denominator margin before accepting arithmetic.

[WARN] WATCH FOR

- Reversing the direction and reporting 45/50 for condition given positive.
- Using all 1,000 screened students as the denominator after a positive result is known.
- Swapping the labels on the 45 true positives and 95 false positives.

SCORING PART (c) — E / P / I

Expected elements

- **rejects-ninety-percent-claim** (required) — States that a positive result does not imply a 90 percent conditional probability in this population
- **computes-positive-conditioned** (required) — Uses 45/140, approximately 32.1 percent, for condition given positive
- **uses-two-positive-counts** (required) — Cites 45 true positives and 95 false positives as the composition of the 140 positive results
- **diagnoses-direction** (required) — Explains that sensitivity is positive given condition, while the claim asks for condition given positive, with different denominators

E	Rejects the claim, reports 45/140 or about 32.1 percent, uses the 45-versus-95 composition, and clearly explains the reversal of the conditioning event and denominator.
P	Gets the 32.1 percent result and rejects the claim but incompletely explains either the two positive counts, the base-rate mechanism, or the change in denominator.
I	Reports 90 percent for condition given positive, uses 45/1,000, or reverses the condition without identifying which group supplies the denominator.

Common mistakes

- Using the sensitivity 45/50 as the chance of condition after a positive result
- Dividing 45 by all 1,000 screened students instead of by the 140 positive tests
- Calling the 95 positives among students without the condition true positives
- Ignoring the 5 percent prevalence when explaining why the two conditional probabilities differ

[STAR] TODAY'S PROBLEM — annotated

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

Mina: "The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it."

Luis: "The base rate matters; the denominator should be everyone who tested positive."

Screening results (counts)

Status	Pos.	Neg.	Total
Condition	45	5	50
No condition	95	855	950
Total	140	860	1,000

FIRST TAKE (5 min)

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

Key: Not graded. The familiar 90 percent number is intentionally tempting; the revision should use the full positive-test group.

Rules callout "THE GIVEN EVENT SETS THE DENOMINATOR (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** For $P(\text{condition} \mid \text{positive})$, identify the numerator count and the denominator count from the table.

Key: The numerator is 45 students who both have the condition and tested positive. The denominator is all $45 + 95 = 140$ students who tested positive.

DOK 2 **(b)** Compute $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$. Show the count fraction for each and convert each to a percent.

Key: $P(\text{positive} \mid \text{condition}) = 45/50 = 0.90 = 90\%$. In the other direction, $P(\text{condition} \mid \text{positive}) = 45/140 = 9/28 \approx 0.321 = 32.1\%$.

DOK 3 **(c)** Decide whether Mina's claim is warranted. Cite the 45 positive results among students with the condition and the 95 positive results among students without it, explain how those counts determine the positive-test group, and diagnose the confusion between $P(\text{positive} \mid \text{condition})$ and $P(\text{condition} \mid \text{positive})$.

Key: Mina's claim is not warranted. The 90% sensitivity is $P(\text{positive} \mid \text{condition})$: it uses the 50 students with the condition as its denominator. After a positive result, the relevant group is instead all 140 positive tests. That group contains 45 true positives and 95 false positives, so $P(\text{condition} \mid \text{positive}) = 45/140 \approx 32.1\%$, not 90%. Because only 50 of 1,000 students have the condition while 950 do not, even a 10% false-positive rate produces more false positives than true positives. Reversing the words after the conditioning bar changes the denominator and answers a different question.

EXIT

One thing the video changed about my first take: ______________________